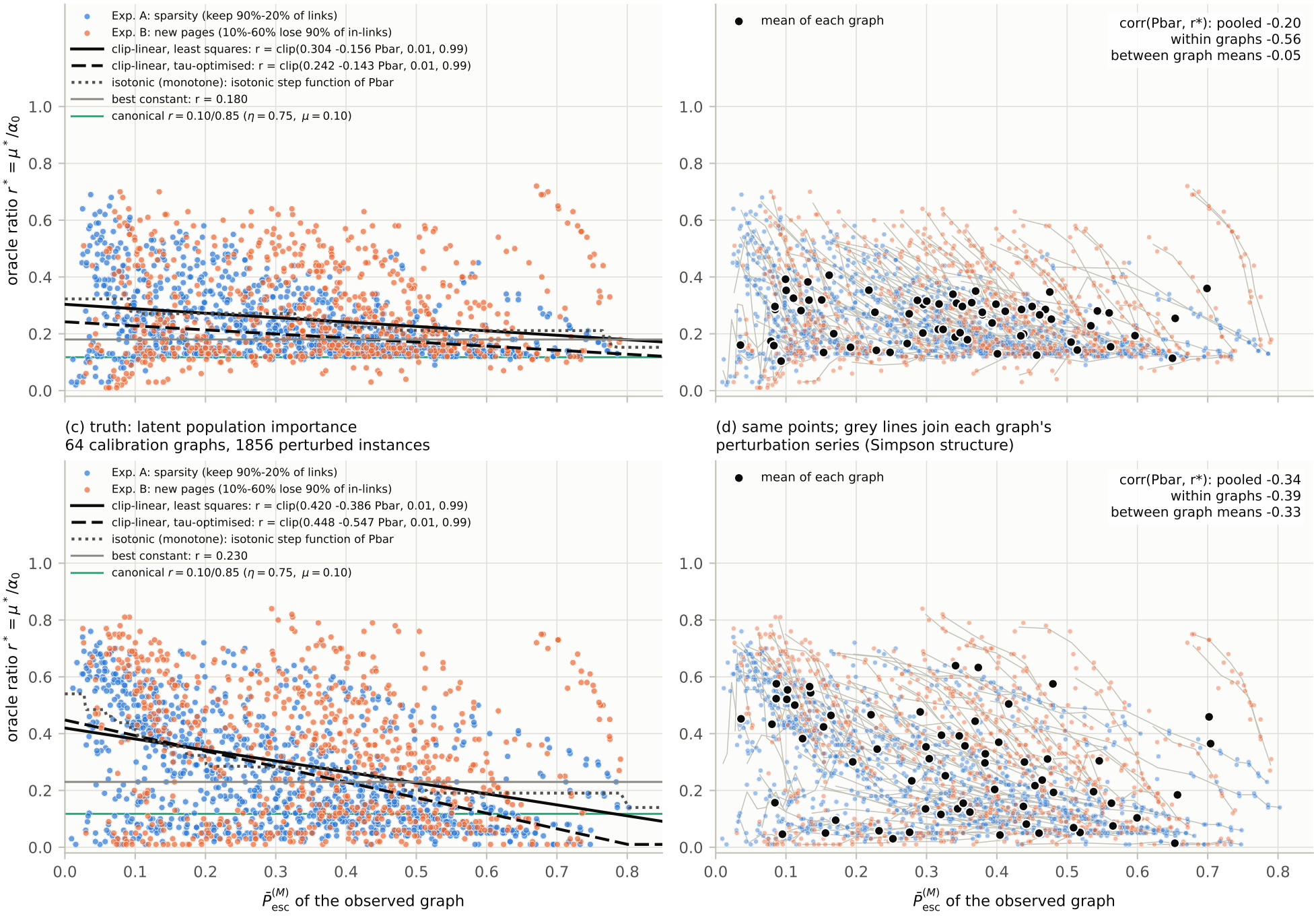

(a) truth: canonical NCDawareRank on the complete graph
64 calibration graphs, 1792 perturbed instances
Exp. A: sparsity (keep 90%-20% of links)
Exp. B: new pages (10%-60% lose 90% of in-links)
clip-linear, least squares: r = clip(0.304 -0.156 Pbar, 0.01, 0.99)
clip-linear, tau-optimised: r = clip(0.242 -0.143 Pbar, 0.01, 0.99)
isotonic (monotone): isotonic step function of Pbar
best constant: r = 0.180
canonical $r = 0.10/0.85$ ($\eta = 0.75$, $\mu = 0.10$)
oracle ratio $r^* = \mu^*/\alpha_0$
(b) same points; grey lines join each graph's
perturbation series (Simpson structure)
mean of each graph
corr(Pbar, r*): pooled -0.20
within graphs -0.56
between graph means -0.05
(c) truth: latent population importance
64 calibration graphs, 1856 perturbed instances
Exp. A: sparsity (keep 90%-20% of links)
Exp. B: new pages (10%-60% lose 90% of in-links)
clip-linear, least squares: r = clip(0.420 -0.386 Pbar, 0.01, 0.99)
clip-linear, tau-optimised: r = clip(0.448 -0.547 Pbar, 0.01, 0.99)
isotonic (monotone): isotonic step function of Pbar
best constant: r = 0.230
canonical $r = 0.10/0.85$ ($\eta = 0.75$, $\mu = 0.10$)
oracle ratio $r^* = \mu^*/\alpha_0$
(d) same points; grey lines join each graph's
perturbation series (Simpson structure)
mean of each graph
corr(Pbar, r*): pooled -0.34
within graphs -0.39
between graph means -0.33
$\bar{P}^{(M)}_{\text{esc}}$ of the observed graph
$\bar{P}^{(M)}_{\text{esc}}$ of the observed graph